American Computer Science League

2025-2026 ● Contest 2: Short Problems ● Senior Division

1. Prefix/Infix/Postfix Notation

What is the value of *x* that makes the following postfix expression evaluate to 2026?

x x 1 + + x 2 + + x 3 + +

A. 404
B. 505
C. 202
D. 1010
E. 2020

2. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

6 4 + 2 ^ 6 3 / / 8 4 / 4 8 / / +

A. 5
B. 50
C. 54
D. 104
E. 204

3. Bit-String Flicking

Evaluate and express the answer in octal:

375_8 AND $A4_{16}$ OR 01110110_2

A. 244
B. 104
C. 266
D. 366
E. 156

4. Bit-String Flicking

Solve the following equation for X if X is a 5-bit string:

(NOT 01100) OR (LCIRC-1 01101 AND X) = X

A. 1*011
B. 10*11
C. *0010
D. 110*1
E. 1111*

5. LISP Evaluate the following LISP expression: (MULT 3 (SUB 8 6) (ADD 2 1 4) (DIV 6 2) (SQUARE 2))	A. 60 B. 126 C. 252 D. 420 E. 504
6. LISP Evaluate the following LISP expression: (SETQ V '(r (b y) (o (p (g))))) (CAR (CDR (CAR (REVERSE (CDR V)))))	A. ((g)) B. (b y) C. (o) D. (p (g)) E. (o (p (g)))

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1. 前缀/中缀/后缀表示法

若以下后缀表达式的计算结果为 2026，求 x 的值：

x x 1 + + x 2 + + x 3 + +

A. 404
B. 505
C. 202
D. 1010
E. 2020

2. 前缀/中缀/后缀表示法

计算以下后缀表达式的值（所有出现的数字均为单个数字）：

6 4 + 2 ^ 6 3 / / 8 4 / 4 8 / / +

A. 5
B. 50
C. 54
D. 104
E. 204

3. 位运算

计算以下表达式，并将结果用八进制（Octal）表示：

375_8 AND $A4_{16}$ OR 01110110_2

A. 244
B. 104
C. 266
D. 366
E. 156

4. 位运算

已知 X 为一个 5 位二进制串（5-bit string），解方程求 X：

(NOT 01100) OR (LCIRC-1 01101 AND X) = X

A. 1*011
B. 10*11
C. *0010
D. 110*1
E. 1111*

5. LISP 对下列 LISP 表达式进行求值： (MULT 3 (SUB 8 6) (ADD 2 1 4) (DIV 6 2) (SQUARE 2))	A. 60 B. 126 C. 252 D. 420 E. 504
6. LISP 对下列 LISP 表达式进行求值： (SETQ V '(r (b y) (o (p (g))))) (CAR (CDR (CAR (REVERSE (CDR V)))))	A. ((g)) B. (b y) C. (o) D. (p (g)) E. (o (p (g)))